

5291 project

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Violation Group vs Borough

```
df <- read.csv("C:/Users/Jack Shang/Downloads/inspection_results_with_ntas.csv")
df$violation_group <- ifelse(
  df$violation_category %in% c(
    "COOLING & REFRIGERATION",
    "HOT HOLDING",
    "COLD HOLDING",
    "TEMPERATURE REGULATING",
    "COOKING",
    "REHEATING & HOT HOLDING"
  ),
  "Temperature Control",

  ifelse(
    df$violation_category %in% c(
      "PEST CONTROL"
    ),
    "Pest Control",

    ifelse(
      df$violation_category %in% c(
        "FOOD PROTECTION",
        "CONTAMINATION",
        "ADULTERATED",
        "UNAPPROVED SOURCE",
        "REDUCE OXYGEN PACKAGE"
      ),
      "Food Protection & Contamination",

      ifelse(
        df$violation_category %in% c(
          "HANDWASH/TOILET",
          "PLUMBING",
          "UTENSILS",
          "WAREWASHING",
          "EQUIPMENT",
          "FACILITY",
          "MAINTENANCE, CONSTRUCTION & PLACEMENT",
          "LIGHT, HEAT & VENTILATION"
        )
```



```
##                2451                42584
##      Temperature Control      Worker Practices
##                17917                2925
```

```
tab <- table(df$borough, df$violation_group)
prop_tab <- prop.table(tab, margin = 1)
prop_tab
```

```
##
##      Food Protection & Contamination Other / Misc
##  Bronx                0.08083026    0.05817783
##  Brooklyn            0.08699187    0.06686992
##  Manhattan           0.09178351    0.07505862
##  Queens              0.10457378    0.07337770
##  Staten Island       0.08552803    0.04250326
##
##      Permit, Signs & Labeling Pest Control Regulatory / Policy
##  Bronx                0.03352959    0.25217044    0.02454845
##  Brooklyn            0.03119919    0.24146341    0.01897019
##  Manhattan           0.02296980    0.21842848    0.02411829
##  Queens              0.03052709    0.22735115    0.01982325
##  Staten Island       0.03233377    0.20651890    0.01929596
##
##      Sanitation & Facilities Temperature Control Worker Practices
##  Bronx                0.36144097    0.16146093    0.02784153
##  Brooklyn            0.36514228    0.16456640    0.02479675
##  Manhattan           0.39218548    0.14920802    0.02624779
##  Queens              0.36276892    0.15862118    0.02295694
##  Staten Island       0.38826597    0.18252934    0.04302477
```

```
chisq_res <- chisq.test(tab)
chisq_res
```

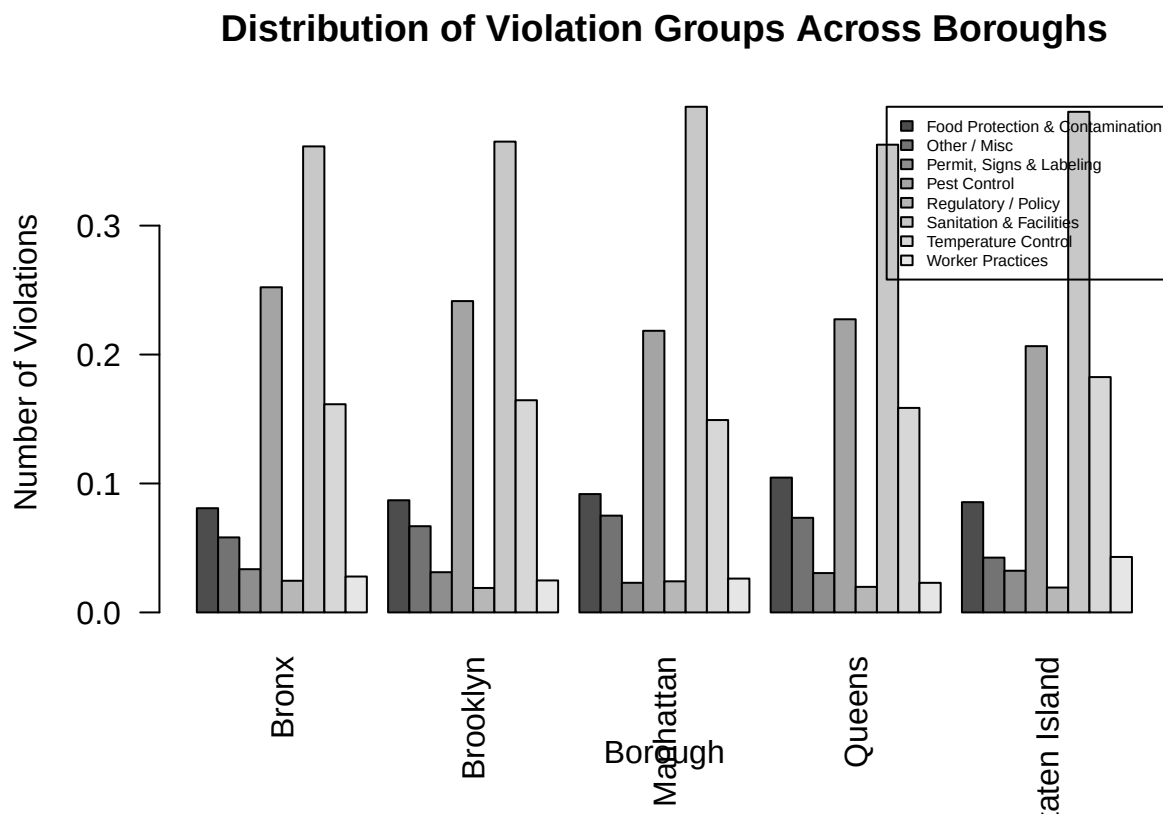
```
##
##  Pearson's Chi-squared test
##
## data:  tab
## X-squared = 486.91, df = 28, p-value < 2.2e-16
```

```
chisq_res$expected
```

```
##
##      Food Protection & Contamination Other / Misc
##  Bronx                927.5321    700.6785
##  Brooklyn            2732.3370    2064.0685
##  Manhattan           3868.4042    2922.2791
##  Queens              2628.7636    1985.8268
##  Staten Island       354.9632    268.1471
##
##      Permit, Signs & Labeling Pest Control Regulatory / Policy
##  Bronx                283.0597    2297.0362    216.26534
##  Brooklyn            833.8410    6766.6408    637.07742
```

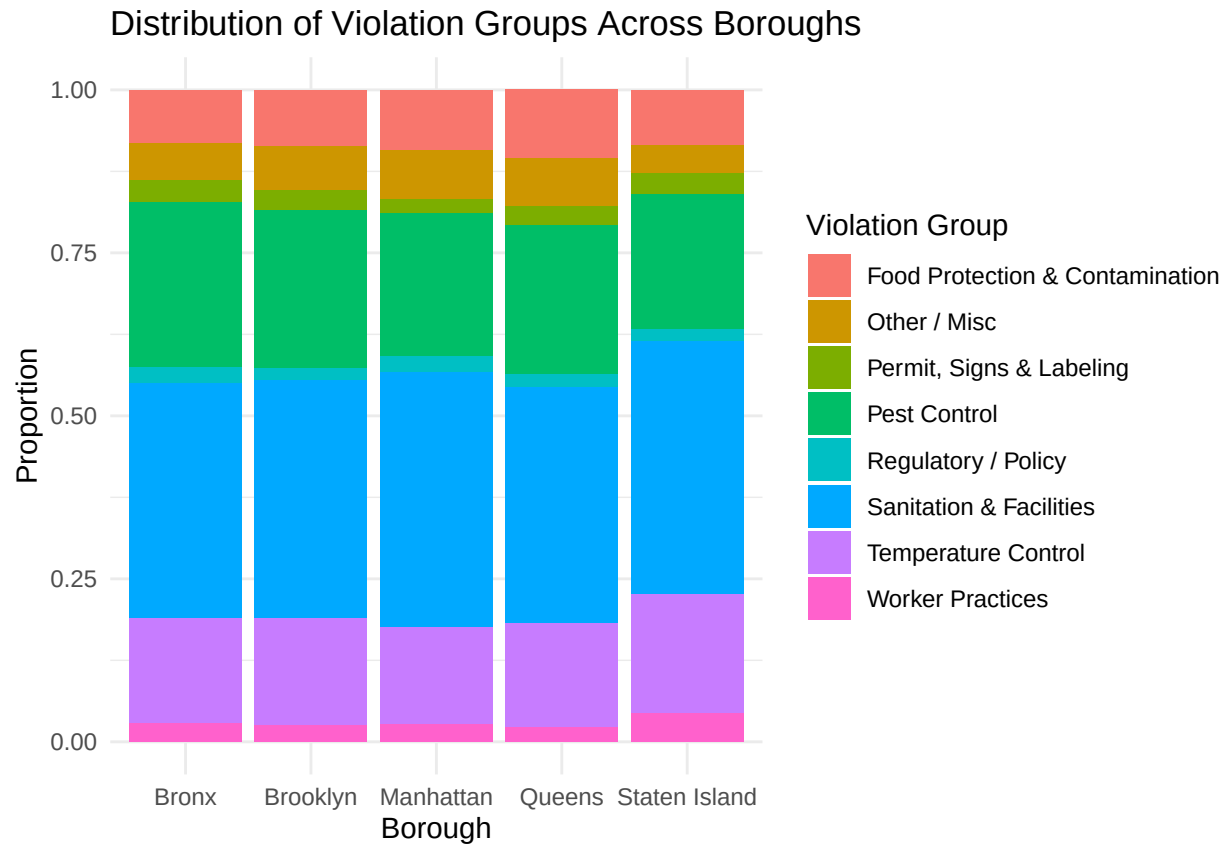
```
## Manhattan 1180.5404 9580.1147 901.96524
## Queens 802.2330 6510.1411 612.92805
## Staten Island 108.3259 879.0673 82.76395
##
## Sanitation & Facilities Temperature Control Worker Practices
## Bronx 3757.423 1580.9164 258.08899
## Brooklyn 11068.668 4657.0853 760.28211
## Manhattan 15670.864 6593.4358 1076.39670
## Queens 10649.093 4480.5515 731.46248
## Staten Island 1437.952 605.0109 98.76971
```

```
barplot(
  t(prop_tab),
  beside = TRUE,
  legend.text = TRUE,
  args.legend = list(x = "topright", cex = 0.5),
  xlab = "Borough",
  ylab = "Number of Violations",
  main = "Distribution of Violation Groups Across Boroughs",
  las = 2
)
```



```
library(ggplot2)
ggplot(df, aes(x = borough, fill = violation_group)) +
  geom_bar(position = "fill") +
```

```
labs(
  title = "Distribution of Violation Groups Across Boroughs",
  x = "Borough",
  y = "Proportion",
  fill = "Violation Group"
) +
theme_minimal()
```

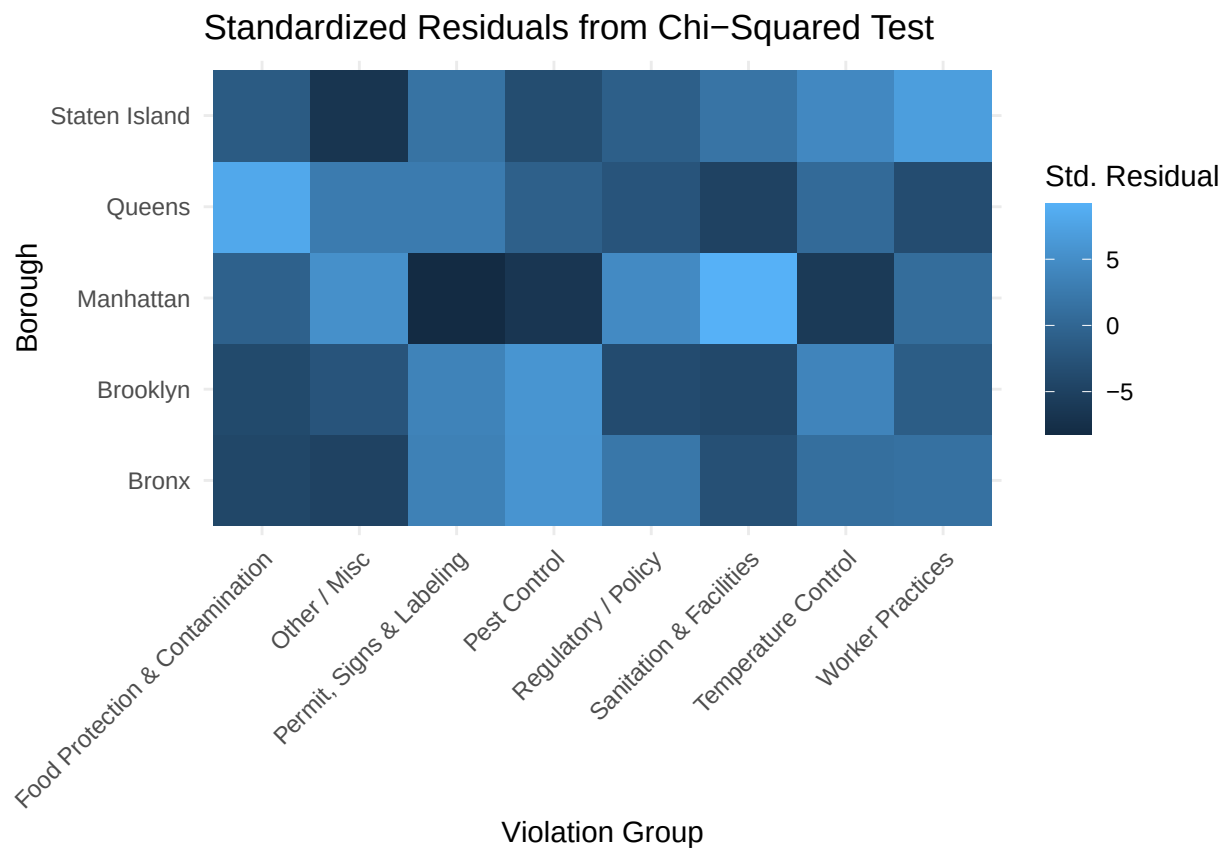


```
chisq_res$stdres
```

```
##
##      Food Protection & Contamination Other / Misc
## Bronx      -4.2426927      -4.8276589
## Brooklyn   -3.8363758      -2.3895333
## Manhattan  -0.6879654       5.1807751
## Queens      8.0679042       2.6378648
## Staten Island -1.5283715     -6.7734513
##
##      Permit, Signs & Labeling Pest Control Regulatory / Policy
## Bronx      3.3429389      5.7236061      2.1407492
## Brooklyn   3.5592189      5.8163466     -3.5886489
## Manhattan  -8.1905064     -6.6035592      4.4898523
## Queens      2.6786480     -0.8662855     -2.3543326
## Staten Island 1.5541674     -3.4028179     -0.9907777
##
```

```
##           Sanitation & Facilities Temperature Control Worker Practices
##  Bronx           -2.9265080           1.0643102           1.3810670
##  Brooklyn        -4.0481748           3.7290564          -1.2079602
##  Manhattan         9.1527887          -6.0334455           0.8003086
##  Queens           -4.8985967           0.4595774          -3.4373388
##  Staten Island     1.7322498           4.2808834           6.8686245
```

```
std_res <- as.data.frame(chisq_res$stdres)
colnames(std_res) <- c("boro", "violation_group", "std_residual")
ggplot(std_res, aes(x = violation_group, y = boro, fill = std_residual)) +
  geom_tile() +
  labs(
    title = "Standardized Residuals from Chi-Squared Test",
    x = "Violation Group",
    y = "Borough",
    fill = "Std. Residual"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
chi2 <- chisq_res$statistic
n <- sum(tab)
k <- min(nrow(tab), ncol(tab))
sqrt(chi2 / (n * (k - 1)))
```

```
## X-squared
```

```
## 0.03273865
```

```
percent <- prop.table(tab, margin = 1) * 100
range_diff <- apply(percent, 2, function(x) max(x) - min(x))
sort(range_diff, decreasing = TRUE)
```

##	Pest Control	Temperature Control
##	4.5651537	3.3321315
##	Other / Misc	Sanitation & Facilities
##	3.2555361	3.0744507
##	Food Protection & Contamination	Worker Practices
##	2.3743526	2.0067834
##	Permit, Signs & Labeling	Regulatory / Policy
##	1.0559784	0.5578259

I first recode the original detailed violation categories into 8 broader violation groups to make the analysis easier to interpret. Overall, Sanitation & Facilities is the most common violation group, followed by Pest Control and Temperature Control. Then I examine whether the distribution of violation groups differ across boroughs. With a chi-squared test, the result is statistically significant, suggesting that violation group distributions are not identical across boroughs. However, The plots show that the overall distribution is broadly similar across boroughs. Hence, I run the Cramer's V, with a quite small number, concluding the difference across boroughs is limited and a weak association, which implies the statistical significance is likely driven partly by the large sample size rather than large differences in violation patterns. Also, standardized residuals and heatmap are used to identify which borough-violation group combinations contributed most to the chi-squared result. On the heatmap, more positive colors indicate higher-than-expected counts, and vice versa.

Violation Group vs Cuisine

```
library(dplyr)
```

```
##
## 'dplyr'

## The following objects are masked from 'package:stats':
##
## filter, lag

## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union
```

```
hypercategory <- c(
  "American"           = "American",
  "New American"       = "American",
  "Californian"        = "American",
  "Steakhouse"         = "American",
  "Barbecue"           = "American",
  "Hamburgers"         = "American",
```

"Soul Food"	= "American",
"Cajun"	= "American",
"Creole"	= "American",
"Creole/Cajun"	= "American",
"Tex-Mex"	= "American",
"Southwestern"	= "American",
"Chinese"	= "East Asian",
"Japanese"	= "East Asian",
"Korean"	= "East Asian",
"Chinese/Japanese"	= "East Asian",
"Chinese/Cuban"	= "East Asian",
"Indian"	= "South & Southeast Asian",
"Thai"	= "South & Southeast Asian",
"Southeast Asian"	= "South & Southeast Asian",
"Bangladeshi"	= "South & Southeast Asian",
"Pakistani"	= "South & Southeast Asian",
"Filipino"	= "South & Southeast Asian",
"Indonesian"	= "South & Southeast Asian",
"Afghan"	= "South & Southeast Asian",
"Asian/Asian Fusion"	= "South & Southeast Asian",
"Mexican"	= "Latin American",
"Latin American"	= "Latin American",
"Caribbean"	= "Latin American",
"Peruvian"	= "Latin American",
"Brazilian"	= "Latin American",
"Chilean"	= "Latin American",
"Polynesian"	= "Latin American",
"Hawaiian"	= "Latin American",
"Italian"	= "European",
"French"	= "European",
"Spanish"	= "European",
"English"	= "European",
"German"	= "European",
"Portuguese"	= "European",
"Polish"	= "European",
"Russian"	= "European",
"Eastern European"	= "European",
"Irish"	= "European",
"Scandinavian"	= "European",
"Czech"	= "European",
"Basque"	= "European",
"New French"	= "European",
"Continental"	= "European",
"Haute Cuisine"	= "European",
"Australian"	= "European",
"Mediterranean"	= "Mediterranean",
"Greek"	= "Mediterranean",
"Turkish"	= "Mediterranean",


```

"Lebanese"           = "Mediterranean",
"Moroccan"          = "Mediterranean",
"Tapas"              = "Mediterranean",

"Middle Eastern"     = "Middle Eastern",
"Egyptian"           = "Middle Eastern",
"Iranian"            = "Middle Eastern",
"Armenian"           = "Middle Eastern",
"Jewish/Kosher"      = "Middle Eastern",

"African"            = "African",
"Ethiopian"          = "African",

"Pizza"              = "Dish-Type",
"Sandwiches"         = "Dish-Type",
"Chicken"            = "Dish-Type",
"Seafood"            = "Dish-Type",
"Hotdogs"            = "Dish-Type",
"Hotdogs/Pretzels"   = "Dish-Type",
"Bagels/Pretzels"    = "Dish-Type",
"Pancakes/Waffles"   = "Dish-Type",
"Salads"             = "Dish-Type",
"Soups"              = "Dish-Type",

"Coffee/Tea"         = "Beverages & Sweets",
"Juice, Smoothies, Fruit Salads" = "Beverages & Sweets",
"Bottled Beverages"  = "Beverages & Sweets",
"Donuts"             = "Beverages & Sweets",
"Frozen Desserts"    = "Beverages & Sweets",
"Bakery Products/Desserts" = "Beverages & Sweets",
"Nuts/Confectionary" = "Beverages & Sweets",

"Vegan"              = "Dietary Style",
"Vegetarian"         = "Dietary Style",
"Fruits/Vegetables"  = "Dietary Style",

"Fusion"             = "Mixed/Uncategorized",
"Sandwiches/Salads/Mixed Buffet" = "Mixed/Uncategorized",
"Soups/Salads/Sandwiches" = "Mixed/Uncategorized",
"Other"              = "Mixed/Uncategorized",
"Not Listed/Not Applicable" = "Mixed/Uncategorized",
" "                  = "Mixed/Uncategorized"
)

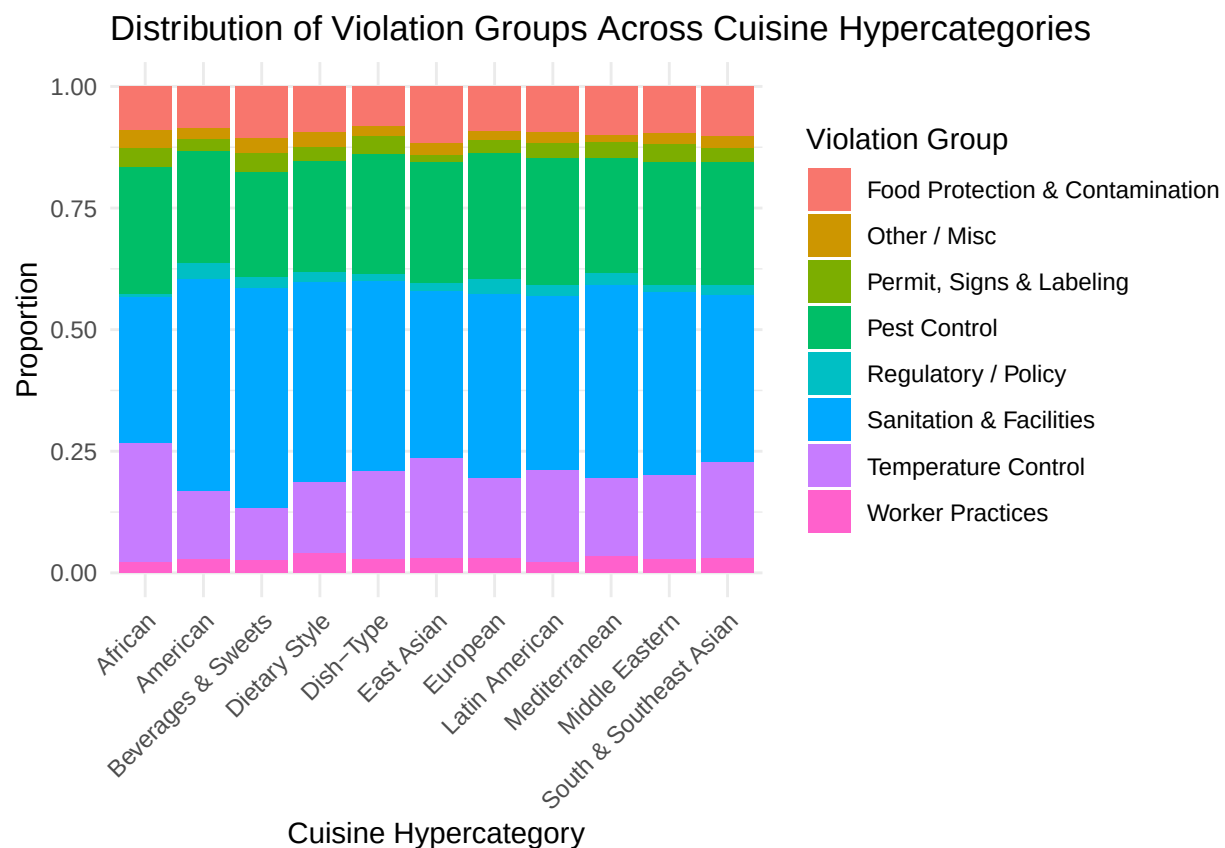
df$hypercategory <- hypercategory[df$cuisine]
df <- df[!is.na(df$hypercategory), ]
df <- df[!is.na(df$score), ]
df <- df %>% filter(hypercategory != "Mixed/Uncategorized")

tab_hyper <- table(df$hypercategory, df$violation_group)
prop_hyper <- prop.table(tab_hyper, margin = 1)
chisq_hyper <- chisq.test(tab_hyper)
chisq_hyper

```

```
##
## Pearson's Chi-squared test
##
## data:  tab_hyper
## X-squared = 2014.8, df = 70, p-value < 2.2e-16
```

```
ggplot(df, aes(x = hypercategory, fill = violation_group)) +
  geom_bar(position = "fill") +
  labs(
    title = "Distribution of Violation Groups Across Cuisine Hypercategories",
    x = "Cuisine Hypercategory",
    y = "Proportion",
    fill = "Violation Group"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

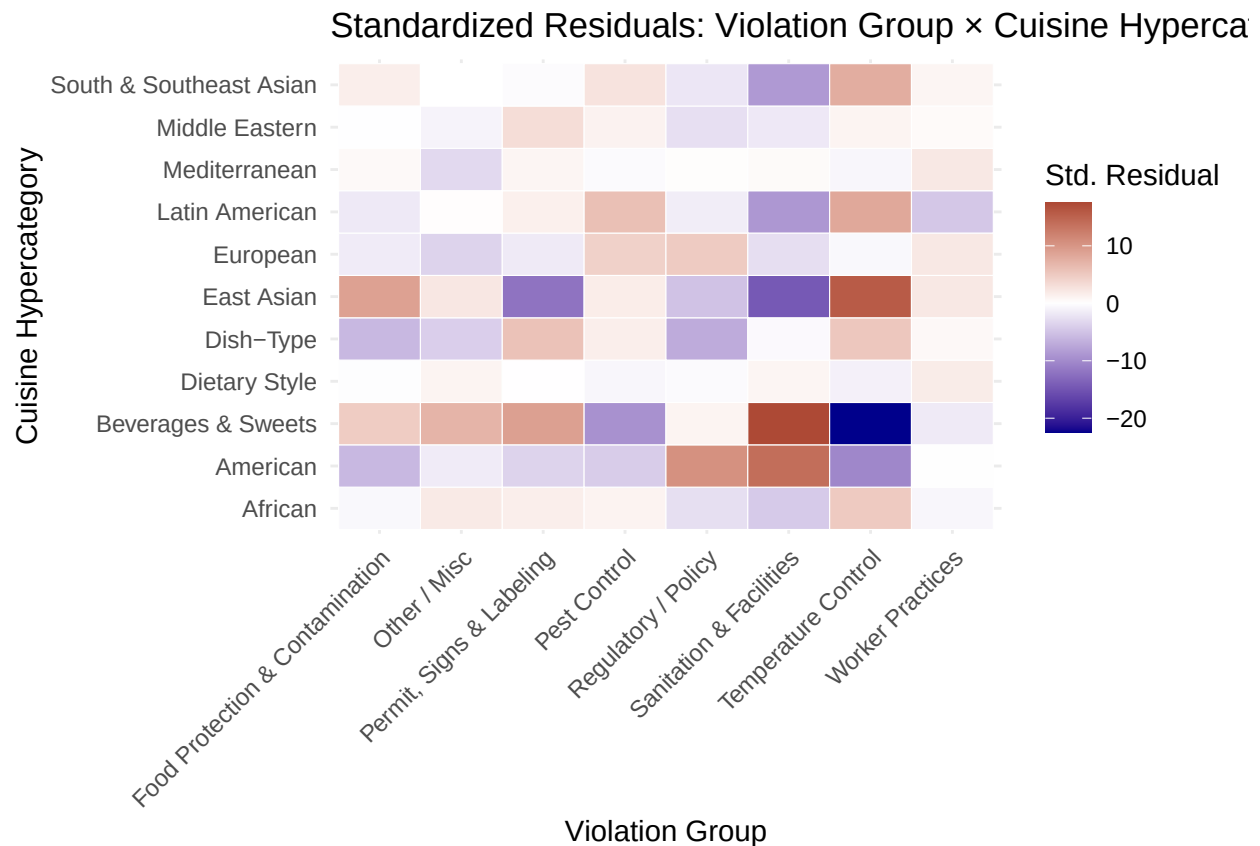


```
std_res_hyper <- as.data.frame(chisq_hyper$stdres)
colnames(std_res_hyper) <- c(
  "hypercategory",
  "violation_group",
  "std_residual"
)
ggplot(std_res_hyper, aes(
  x = violation_group,
```

```

y = hypercategory,
fill = std_residual
)) +
geom_tile(color = "white") +
scale_fill_gradient2(
  low = "darkblue",
  mid = "white",
  high = "darkred",
  midpoint = 0
) +
labs(
  title = "Standardized Residuals: Violation Group × Cuisine Hypercategory",
  x = "Violation Group",
  y = "Cuisine Hypercategory",
  fill = "Std. Residual"
) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



```

n <- sum(tab_hyper)
chi2 <- chisq_hyper$statistic
r <- nrow(tab_hyper)
c <- ncol(tab_hyper)
sqrt(chi2 / (n * min(r - 1, c - 1)))

```

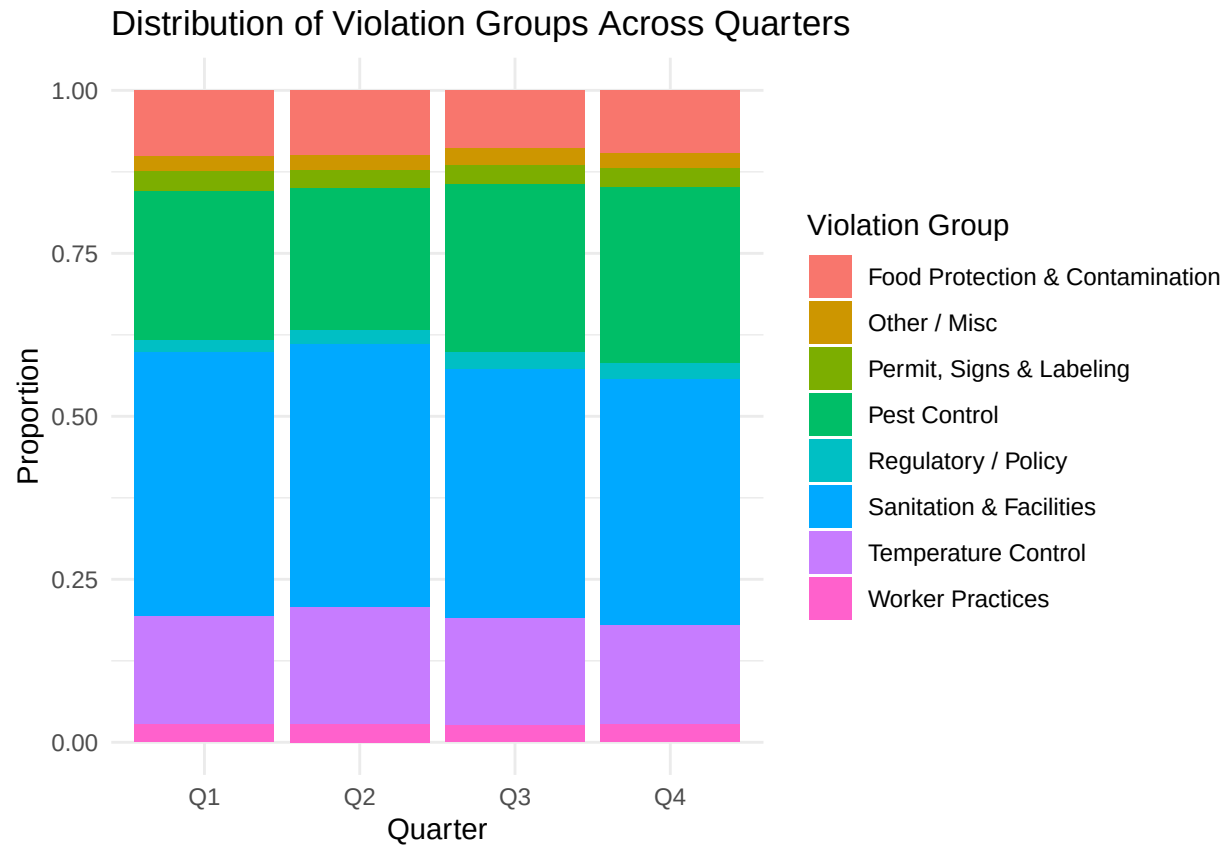
```
## X-squared
## 0.05209183
```

Violation Group vs Quarter

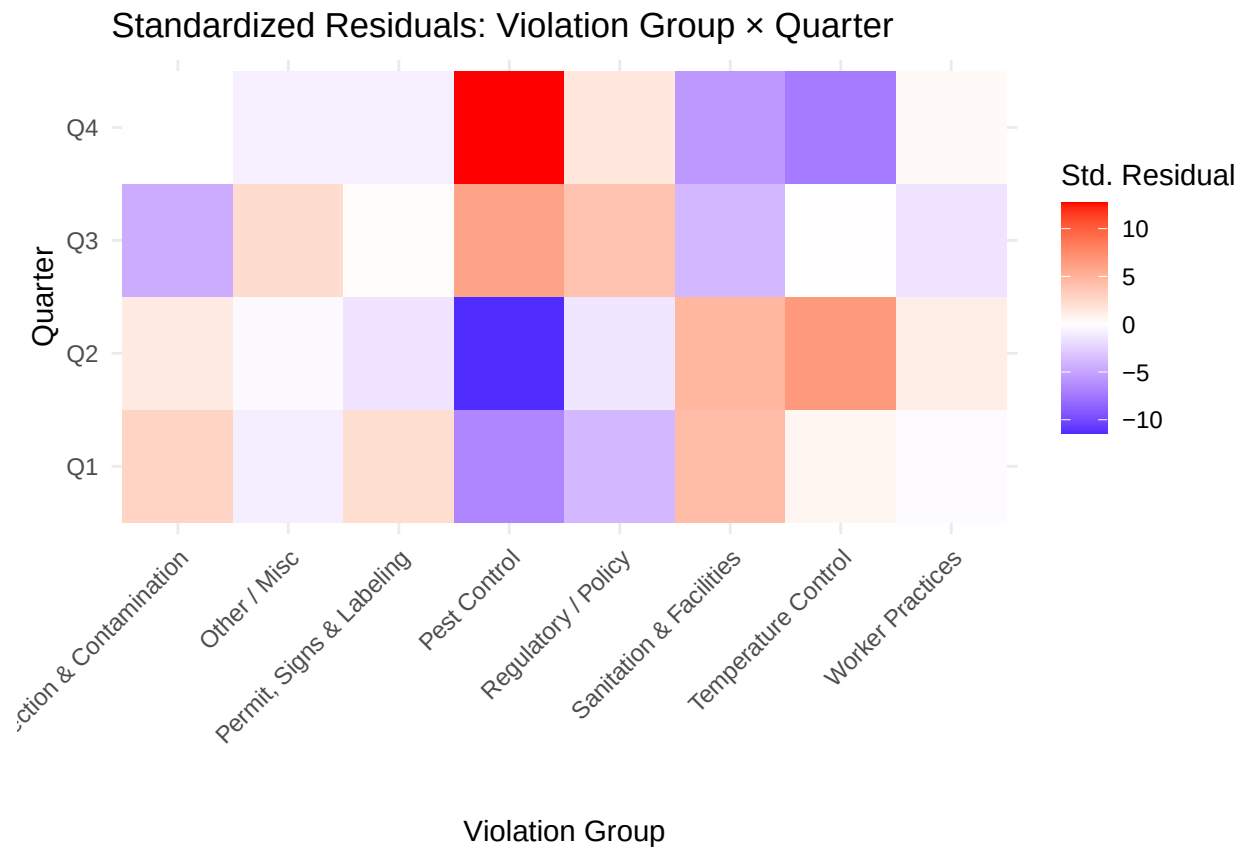
```
df$inspection_date <- as.Date(df$inspection_date, format = "%m/%d/%Y")
df$quarter <- paste0(
  "Q",
  ceiling(as.numeric(format(df$inspection_date, "%m")) / 3)
)
df_qt <- df[
  !is.na(df$violation_group) &
  !is.na(df$quarter),
]
tab_qt <- table(df_qt$quarter, df_qt$violation_group)
chisq_qt <- chisq.test(tab_qt)
chisq_qt
```

```
##
## Pearson's Chi-squared test
##
## data:  tab_qt
## X-squared = 368.85, df = 21, p-value < 2.2e-16
```

```
ggplot(df, aes(x = quarter, fill = violation_group)) +
  geom_bar(position = "fill") +
  labs(
    title = "Distribution of Violation Groups Across Quarters",
    x = "Quarter",
    y = "Proportion",
    fill = "Violation Group"
  ) +
  theme_minimal()
```



```
std_res_qt <- as.data.frame(chisq_qt$stdres)
colnames(std_res_qt) <- c(
  "quarter",
  "violation_group",
  "std_residual"
)
ggplot(std_res_qt, aes(x = violation_group, y = quarter, fill = std_residual)) +
  geom_tile() +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
    midpoint = 0
  ) +
  labs(
    title = "Standardized Residuals: Violation Group × Quarter",
    x = "Violation Group",
    y = "Quarter",
    fill = "Std. Residual"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
n <- sum(tab_qt)
sqrt(chisq_qt$statistic / (n * (min(dim(tab_qt)) - 1)))
```

```
## X-squared
## 0.03404549
```